Final Exam Practice Problems

Computer Graphics, Wolfram Mathematica

```
In[1]:= SetOptions[EvaluationNotebook[], ShowCellLabel → True];

In[2]:= R = 1; P1 = 0; P2 = 0; h = 2.5; Ru = 1.5;

In[3]:= Base = Cylinder[{{0, 0, 0}, {0, 0, 0.25}}, R];

In[4]:= Holder = Arrow[{{0, 0, 0}, {0, 0, h}}];

In[5]:= Points = Table[{Ru * Sin[Pi / 10 * i], Ru * Cos[Pi / 10 * i], h}, {i, 20}];

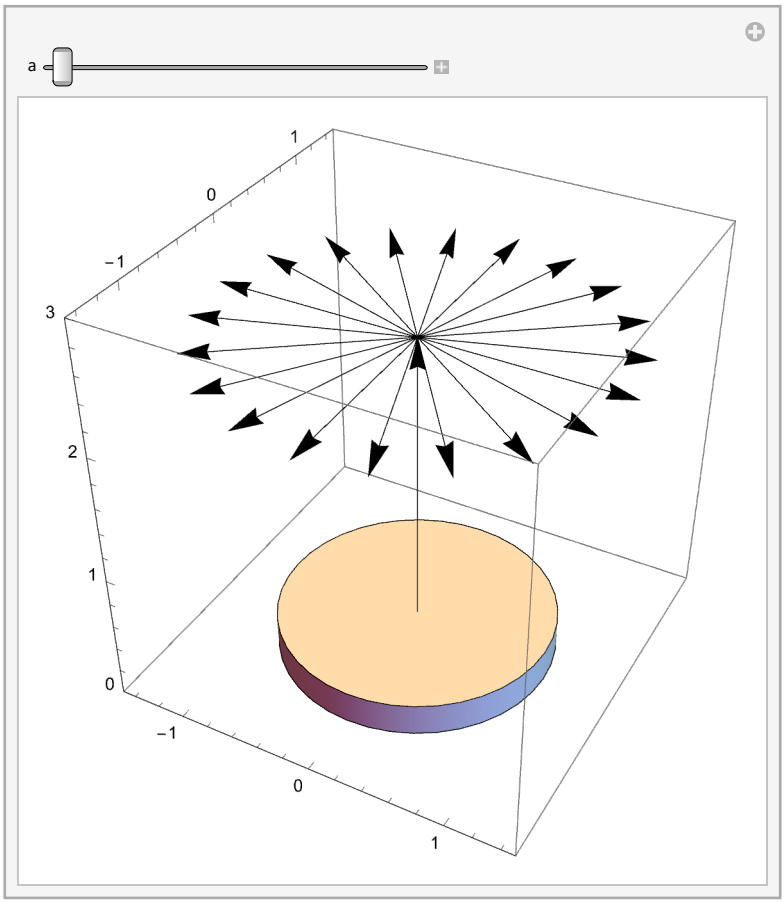
In[6]:= Topp = Table[Arrow[{{0, 0, h}, Points[[i]]}], {i, 20}];

In[7]:= Obj = {Base, Holder, Topp};

In[8]:= GeoObj[a_] := GeometricTransformation[Obj,
          Composition[TranslationTransform[{P1, P2, 0}], RotationTransform[a, {0, 0, 1}]]]
```

In[9]:=
```
Manipulate[Graphics3D[GeoObj[a],
  PlotRange → {{-1.5, 1.5}, {-1.5, 1.5}, {0, 3}}, Axes → True], {a, 0, 2 * Pi}]
```

Out[9]=



In[10]:=
```
(* Problem 3 *)
```

In[11]:=
```
CurveGr = ParametricPlot[{Sin[2 * t], Cos[3 * t]}, {t, 0, 2 * Pi}][[1]];
```

In[12]:=
```
Cir = {Blue, Thick, Circle[{0, 0}, 1.5]};
```

In[13]:=
```
Tri = {RegularPolygon[{0, 0}, 1.5, 3]};
```

In[14]:=
```
Pen = {RegularPolygon[{0, 0}, 1.5, 5]};
```

In[15]:=
```
Lin = {Red, Thick, Line[{{-1, 0}, {1, 0}}]};
```

In[16]:=
```
Menu3 = PopupMenu[Dynamic[m3],
  {1 → "Curve", 2 → "Circle", 3 → "Triangle", 4 → "Pentagon", 5 → "Line101"}]
```

Out[16]=

Curve

In[17]:=
```
ObjList3 = {CurveGr, Cir, Tri, Pen, Lin};
```

In[18]:=
```
GeoObj3[a_] := Dynamic[GeometricTransformation[ObjList3[[m3]],
    Composition[RotationTransform[a], TranslationTransform[{2.5, 0}]]]]
```

In[19]:=
```
{{Menu3, Dynamic[m3]},
 Dynamic[Manipulate[Graphics[GeoObj3[a], PlotRange → 4, Axes → True], {a, 0, 2 * Pi}]]}
```

Out[19]=

